MIL-E-1/1283B(NAVY)
3 March 1975
SUPERSEDING
MIL-E-1/1283A(NAVY)
7 June 1966
(See note 20)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 7471

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, external tuning cavity, 5,500 to 5,800 MHz, waveguide output.

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Irs
Unit:	V	Vdc	Vdc	mAdc
Maximum:	7.0	350	-500	63
Minimum:	5.7	---	-20	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1.
Weight:	13.25 ounces (375.63 grams) maximum.
Base:	Molded leads, 18 inches (475.2 mm) minimum length (see figure 1).
Output flange:	Mates with waveguide shown on figure 2.

TEST CONDITIONS: See note 1

Parameter:	Ef	Ers	Er	Irs	f
Unit:	V	Vdc	Vdc	mAdc	KHz
	6.3	300	Adjust	37.5 ±12.5	F2

Frequency	
F1	5,500 MHz
F2	5,650 MHz
F3	5,800 MHz

GENERAL:

Qualification - Required
Shelf life - See notes 4 and 5

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1016	Moisture resistance	-		---	---	---	---
4280	Electronic tuning range	2,11	F = F1 thru F3	ΔF	30	---	MHz
1011	Humidity	2,6		R Po ΔF	500 30 -7	--- 65 +7	Meg mW MHz
---	Altitude frequency stability	2	0.02 atmosphere to 1.0 atmosphere in 60 seconds maximum	ΔF	---	-1.5	MHz
---	Acceleration frequency stability	2,7,19		ΔF	-0.2	+0.2	MHz
---	Shock	2,8,19		ΔF	-1.0	+1.0	MHz
	Quality conformance inspection, part 1	3					
4229	Total reflector current	-	Tube not oscillating	Ir	---	5	μAdc
1256	Resonator current	2		Irs	25	50	mAdc
1231	Emission current	-	Tube not oscillating, Ef = 6.3 V; Ef = 5.7 V	ΔIk/Ik	---	-15	%
---	Vibration particle test	9,19		Ir	---	10	μA
4223	Mechanical tuning range	2,10		F	5,500	5,800	MHz
4250	Power output	-	Er adjusted for Po (max); F = F1 thru F3	Po	30	65	mW
4213	Reflector voltage	2	F = F1 thru F3	Er Er	-68 ---	--- -126	Vdc Vdc
4280	Electronic tuning range	2,11	F = F1 thru F3	ΔF	30	---	MHz
---	Reflector modulation sensitivity	2,12,18	F = F1 thru F3	Coeff	1	2.5	MHz/V
---	Spurious modes	2,13		---	---	---	---
4231	Electronic hysteresis	2,14		---	---	50	%
	Quality conformance inspection, part 2						
4289	Heater current	-	Ef = 6.3 V	If	1.08	1.32	A
---	Heater voltage sensitivity	2	ΔEf = ±10%; t = 2 minutes	Coeff	-2.5	+2.5	MHz/V
---	Resonator voltage sensitivity	2	ΔErs = ±1%	Coeff	-0.7	+0.7	MHz/V
4225	Residual FM	2,15		ΔF	---	0.1	MHz/p-p
4302	Frequency change (Warmup)	2,16		ΔF	-7.0	+7.0	MHz
---	Vibration frequency stability	2,17,19		ΔF	---	0.2	MHz/p-p
	Quality conformance inspection, part 3						
---	Life test	-	Group D; F = F3; duty cycle, 15 minutes on, 25 minutes off	t	500	---	hrs
---	Life-test end point:						
4250	Power output	-	F = F1 thru F3	Po	20	---	mW

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NOTES:

1. All tests shall be made with the tube rigidly connected to double ridged waveguide shown on figure 2. The load VSWR for the tube shall be 1.2:1 or less. Mode suppressor and iris screws shall remain as factory set.

2. Power output test of quality conformance inspection, part 1, to be performed prior, during, or after this test.

3. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

4. The tube shall be capable of storage for 12 months in dehydrated shipping containers from -54° *to +71° C without impairment of subsequent performance.* The tube shall withstand this period of storage without requiring any maintenance. The tube shall meet the requirements of power output, quality conformance inspection, part 1, after 2 minutes of initial warmup time.

5. Simulation of transportation and storage temperatures shall be as follows:

   (a) Cold cycle. With the tube in the test chamber and beginning at laboratory ambient temperature, the temperature of the chamber shall be cycled from +21° to -54°C and back to +21°C. This cycle shall be repeated 5 times. The -54°C temperature shall be maintained for at least three hours during each cycle. Also, the +21°C temperature shall be maintained for at least three hours between cycles. After completion of the required cycles, the tube shall be stabilized at the ambient laboratory temperature and then tested for satisfactory performance.

   (b) Hot cycle. Starting at laboratory ambient temperature, the temperature of the chamber shall be cycled from +21° to +71°C and back to +21°C. This cycle shall be repeated 5 times. The +71°C temperature shall be maintained for a least 3 hours during each cycle. Also, the +21°C temperature shall be maintained for at least 3 hours between cycles. At the conclusion of the last cycle, the tube shall be stabilized at ambient laboratory temperature and then checked for satisfactory performance. Either the hot or cold cycle may be performed first. The rate of temperature change of the test chamber from +21° to -54°C or +71°C shall be accomplished in no more than 3 hours. Likewise, the change from +71° or -54°C to +21°C shall require no more than 3 hours.

6. Without application of power, the tube shall be exposed to +52°C temperature and 85 to 95 percent relative humidity conditions for a period of 24 hours. At the conclusion of the 24-hour period, and with temperature and humidity conditions maintained, the resistance between reflector and resonator shall not be below the minimum specified herein. After completion of the leakage resistance test all tube voltages shall be applied and the power output and operating frequency monitored for a period from 20 seconds to 1,830 seconds after application of the voltages. The power output and maximum frequency deviation from the reference frequency (fo) shall be within the limits specified herein. In order to prevent the formation of water droplets on the cavity walls due to seepage between the tube and the connecting waveguide flange, a pressurizing gasket may be interposed between the tube and its mating waveguide flange.

7. Centrifugal acceleration. *Acceleration of 60G shall be applied to the tube in all three planes* in both directions, along the three mutually perpendicular axes, as shown on figure 1, for periods of at least 40 seconds each. The frequency deviation shall not exceed the value specified herein when measured before and after acceleration.

8. Thrust shall be applied in both directions along the three mutually perpendicular axis shown on figure 1. The tube shall be subjected to average accelerations of at least 60G with a rise time of 0.003 + 0.001 second and a dwell time or duration of 0.010 ±0.005 second. The duration is defined as the interval of time during which the average acceleration does not fall below 60G. The tube shall be oscillating during the tests and shall return to *within ±1 MHz of reference* frequency (fo) 1 second after the end of the shock.

9. With the tube operating, it shall be vibrated sinusoidally in a plane perpendicular to the beam axis with the cathode end up. The tube shall be subjected to an acceleration of 15G peak at 60 Hz for 120 seconds. The reflector current shall not exceed the value specified herein when measured with a Brush Recorder BL-202, recording instrument, or equivalent, having a 10 millisecond maximum response time.

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NOTES: -Continued

10. There shall be one frequency tuning mechanism (with a friction locking device) capable of accomplishing smooth tuning to the desired frequency without abrupt frequency changes. The tube shall be capable of being tuned within ±1 MHz of any desired point within the range specified herein. The tube shall tune over the mechanical tuning range specified herein on a single reflector voltage mode.

11. In instances where the klystron mode curve is not symmetric, the electronic tuning range shall be evaluated by measuring the frequency excursion to the nearest half power point and then multiplying by two.

12. The reflector modulation sensitivity shall be defined as the ratio of the half power bandwidth divided by the difference in reflector voltage corresponding to the half power points.

13. Any spurious modes which exist shall be outside the frequency range of 5,450 to 5,850 MHz. Under conditions described in note 14(a), any spurious modes which exist shall not interfere with or cause any frequency discontinuities of the operating mode above the 50 percent power points of the operating mode.

14. Starting at the peak of the mode, the reflector voltage shall be decreased until the tube falls out of oscillation. The reflector voltage shall then be increased slowly until the tube oscillates. Power, at the point when the tube starts oscillating, shall be measured. The reflector voltage shall be increased until the tube falls out of oscillation. The reflector voltage shall then be decreased until the tube oscillates. Power output at the point when the tube starts oscillating shall also be measured. Maximum power recorded in either of these measurements divided by the power at the peak of the mode defines the percentage of electronic hysteresis.

    (a) With the reflector voltage adjusted to allow operation above the 95 percent power points on the mode, the reflector voltage shall be increased until the tube oscillates at the 20 percent power point. The reflector voltage shall be decreased until the other 20 percent power point is reached and then readjusted for operation in the 95 percent power region. Under these conditions there shall be no frequency discontinuities except for possible spurious modes above the 20 percent power points.

15. The peak-to-peak deviation shall be as specified herein.

16. The tube shall be adjusted to the reference frequency (fo) after a warmup period of 30 minutes at normal room ambient temperature. The tube shall then be placed in a test chamber and the chamber temperature shall be reduced to -29°C. The tube shall be permitted to stabilize at -29°C for not less than 2 hours. At the conclusion of this period, and with -29°C temperature maintained, the tube shall be tested in accordance with the following activation time:

    (a) The tube shall be tested for power output and frequency deviation from 20 seconds to 1 hour after power is applied.

    (b) This test shall be repeated starting from a stabilized temperature of 100°C.

17. With the tube operating it shall be vibrated sinusiodally along each of three mutually perpendicular axes, as shown on figure 1. The tube shall be vibrated at an acceleration of 10G over the range from 10 to 80 Hz for 30 seconds (below 30 Hz the double amplitude may be held constant at 0.5 inch (12.70 mm)). The tube shall then be vibrated at 10G peak over the range 80 to 1,000 Hz for not less than 60 seconds. The frequency range shall be scanned smoothly so as to change the frequency exponentially, that is, with a constant rate of rotation of a logarithmic control. Frequency deviation shall not exceed the values specified herein. Accelerations are peak values, not peak to peak.

18. The reflector modulation sensitivity at the peak of the mode shall not be less than 0.2 MHz/V. This test shall be measured with a maximum sweep range of ±2.5 MHz.

19. This is a destructive test. Tubes subjected to this test shall not be delivered under contract or order.

20. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

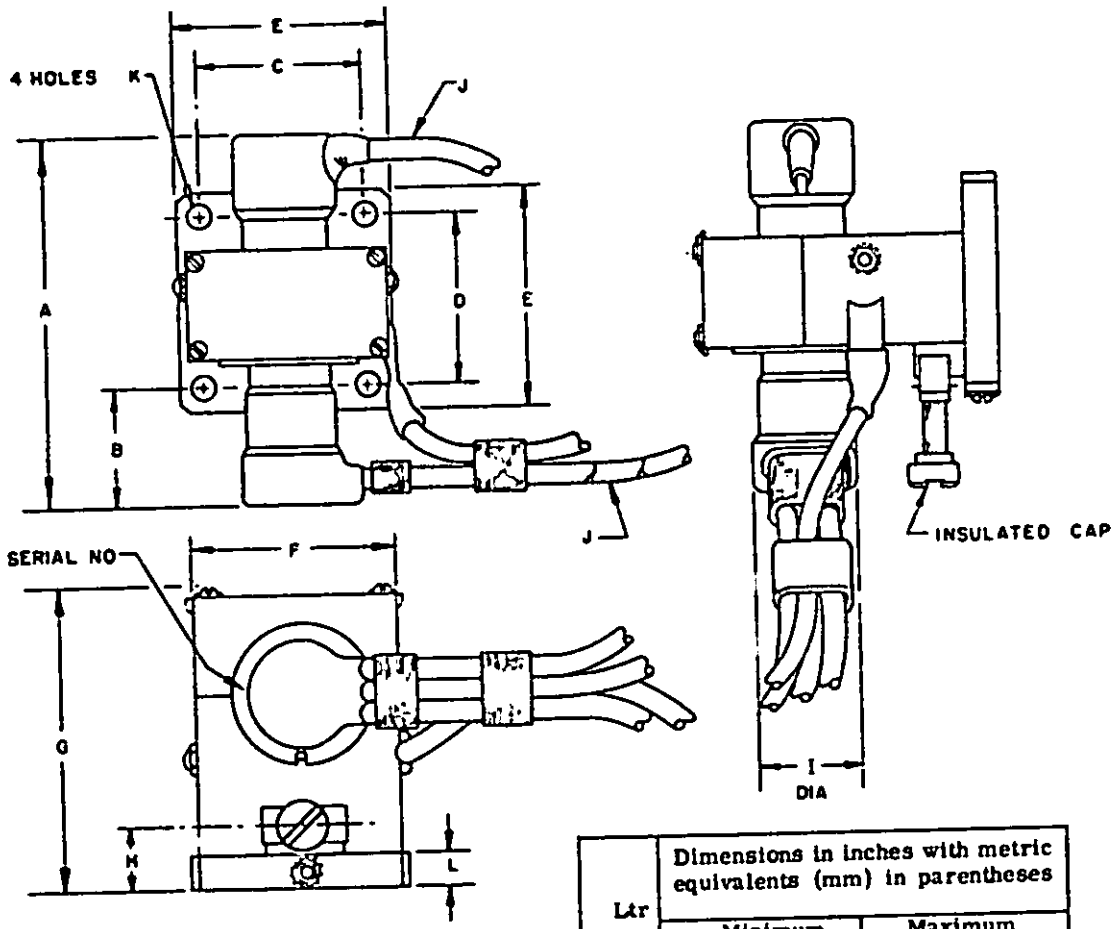
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		3.000 (76.20)
B		.945 (24.00)
C	1.216 (30.89)	1.224 (31.09)
D	1.276 (32.41)	1.284 (32.61)
E	1.620 (41.15)	1.640 (41.66)
G		2.440 (61.98)
H	.440 (11.18)	.560 (14.22)
I		.800 (20.32)
J	17.000 (431.80)	
K	.190 (4.83)	.194 (4.93)
L	.240 (6.10)	.260 (6.60)
Reference dimension		
F	1.560 (39.62) NOM	

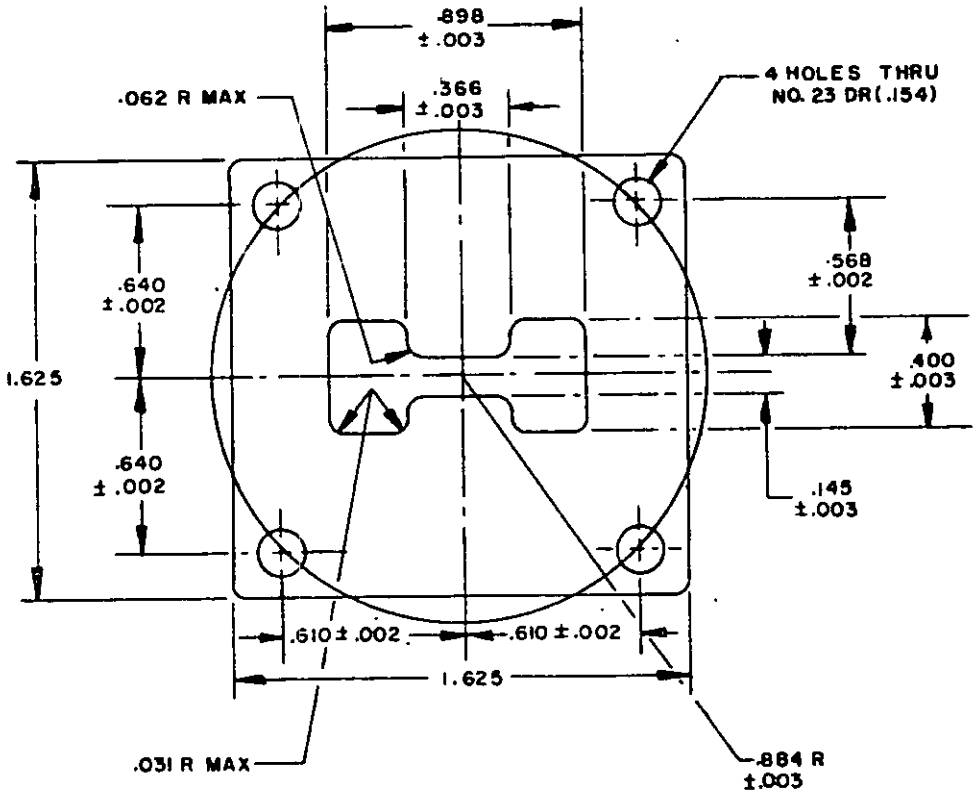
CABLE CONNECTIONS

Yellow - Heater
White - Heater*
Green - Cathode*
Gray - Reflector
Tan - B+

*Internally connected

FIGURE 1. Outline drawing of electron tube type 7471.

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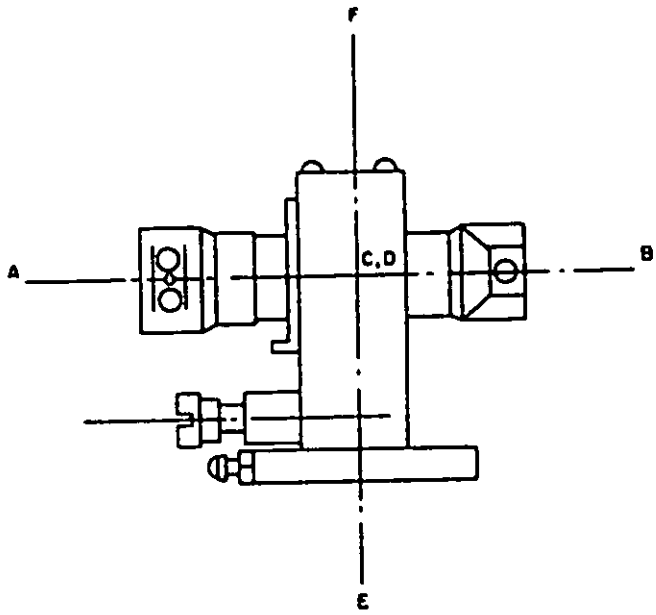


INCHES	MM	INCHES	MM
.002	0.05	.400	10.16
.003	0.08	.568	14.43
.031	0.79	.610	15.49
.062	1.57	.640	16.26
.145	3.68	.884	22.45
.154	3.91	.898	22.81
.366	9.30	1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Double ridged waveguide.

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NOTE:
C, D is in the forward direction into the paper.

FIGURE 3. Acceleration axes for applying thrust.

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